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import tkinter as tk
from tkinter import messagebox, simpledialog
import random

def get_sensor_temperature():
    # Simulate a changing temperature sensor value
    return round(20 + random.uniform(0, 10), 1)

class SmartRoomController:
    def __init__(self, root):
        self.root = root
        root.title("Smart Room Temperature Controller")
        root.geometry("320x250")

        self.current_temp = tk.DoubleVar(value=get_sensor_temperature())
        self.set_temp = tk.DoubleVar(value=24.0)
        self.status = tk.StringVar(value="System Ready")

        tk.Label(root, text="Smart Room Temperature Controller", font=("Arial", 10, "bold")).pack(pady=4)

        tk.Label(root, text="Current Temperature (°C):").pack()
        self.current_temp_label = tk.Label(root, textvariable=self.current_temp)
        self.current_temp_label.pack()

        tk.Label(root, text="Set Temperature (°C):").pack()
        tk.Label(root, textvariable=self.set_temp).pack()

        btn_frame = tk.Frame(root)
        btn_frame.pack(pady=8)

        tk.Button(btn_frame, text="Auto Mode", command=self.auto_mode).pack(side=tk.LEFT, padx=2)
        tk.Button(btn_frame, text="Set from User", command=self.set_temp_user).pack(side=tk.LEFT,
        padx=2)
        tk.Button(root, text="Exit", command=root.quit).pack(pady=2)

        tk.Label(root, text="System Status:").pack()
        tk.Label(root, textvariable=self.status).pack()

        # Start auto-updating the temperature
        self.update_temperature()

    def update_temperature(self):
        temp = get_sensor_temperature()
        self.current_temp.set(temp)
        self.status.set("Sensor value auto-updated")
        self.root.after(1000, self.update_temperature) # Schedule the next update in 1 second

    def auto_mode(self):
        self.set_temp.set(self.current_temp.get())
        self.status.set("Auto mode: matched set temp to current temp")

    def set_temp_user(self):
        try:

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user_temp = simpledialog.askstring("Set Temperature", "Enter desired temperature in °C:")
if user_temp is not None:
    self.set_temp.set(float(user_temp))
    self.status.set("Set temperature updated by user")
else:
    self.status.set("User cancelled input")
except (TypeError, ValueError):
    self.status.set("Invalid input for temperature")

if __name__ == "__main__":
    root = tk.Tk()
    app = SmartRoomController(root)
    root.mainloop()
```